



Wastewater-borne markers of neurodegenerative disease: β -methylamino-*L*-alanine and aminomethylphosphonic acid

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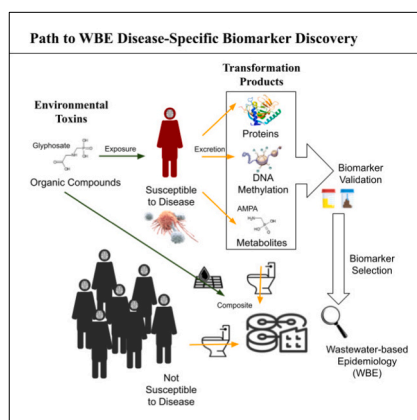
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HIGHLIGHTS

- Detection of environmental toxins associated with neurodegenerative diseases (NDDs)
- LC-MS/MS method developed for β -methylamino-*L*-alanine (BMAA) in wastewater
- Algal bloom reports and U.S. ALS prevalence significantly associated geographically
- Prevalence rates of NDDs Alzheimer's, Parkinson's, and ALS need higher resolution
- Glyphosate's metabolite aminomethylphosphonic acid (AMPA) potential marker of NDDs

GRAPHICAL ABSTRACT



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ABSTRACT

Exposure to toxic organic chemicals such as β -methylamino-*L*-alanine (BMAA) and glyphosate has been associated with neurodegenerative diseases (NDDs), including amyotrophic lateral sclerosis (ALS), Parkinson's Disease (PD), and Alzheimer's Disease (AD). We explored the utility of BMAA and glyphosate's metabolite aminomethylphosphonic acid (AMPA) for serving as potential markers of NDDs by comparing levels of wastewater-borne BMAA and AMPA with regional U.S. rates of NDD prevalence. Newly developed liquid chromatography tandem mass spectrometry (LC-MS/MS) methods were applied to U.S. wastewater samples ($n = 87$) and resultant concentrations of putative biomarkers were statistically compared to NDD prevalence rates in

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conjunction with environmental data on algal blooms and agricultural glyphosate use. Locations of algal blooms were found to be significantly associated ($p = 0.01$) with ALS prevalence rates per 100,000 people. BMAA levels in wastewater were highly correlated ($p < 0.0001$) with ALS prevalence rates by region. BMAA in wastewater typically peaked in summer months. We conclude that NDD biomarker detection in wastewater holds potential value, with BMAA outperforming AMPA. Furthermore, prevalence data for NDDs may have to be reported to the Centers for Disease Control and Prevention at a higher geospatial resolution to further enhance the value for the present type of analysis. Further method development is needed for AMPA to be quantified using LC-MS/MS. Future method developments focusing on metabolites (e.g., AMPA) may enable epidemiologists to determine human exposure levels rather than the mere occurrence of toxic organic chemicals in the environment.

1. Introduction

Individuals living with neurodegenerative diseases (NDDs) suffer from loss of neuronal function, resulting in progressive neuropsychiatric conditions including Alzheimer's Disease (AD), Parkinson's Disease (PD), and amyotrophic lateral sclerosis (ALS), or Lou Gehrig's Disease (Gadhav et al., 2024). The global age-adjusted prevalence of the primary disorder leading to dementia, AD, increased by 1.7 % to 712 per 100,000 people between 1990 and 2016 (Nichols et al., 2019). Global prevalence of PD increased by approximately 160 % to 106 per 100,000 people between 1990 to and 2019 (Ou et al., 2021). Global prevalence of ALS, a motor neuron disease, was reported at 3.3 (2.9, 3.8; 95 % CI) individuals per 100,000 people (Steinmetz et al., 2024); this global age-standardized average, reported with data up to 2021, decreased from the 2018 analysis reporting a prevalence of 4.5 (3.4, 5.6; 95 % CI) per 100,000 people (Logroscino et al., 2018). Nations ranking highest by age-adjusted disability-adjusted life years (DALYs) for motor neuron disease are in the highest income regions of Australia, Asia Pacific, North America, and Western Europe (Steinmetz et al., 2024).

In the United States (US), the prevalence rate (per 100,000 people) of PD is relatively high compared to the other NDDs with an average of 246, with the lowest rates being reported in the states of Alaska and Utah (162 and 164, respectively) and the highest rates reported in the states of Florida and Maine (310 and 300, respectively) (Marras et al., 2018). For AD, the prevalence rates in the United States have averaged 184 per 100,000 people (Alzheimer's, 2023). Similar to PD rates, Alaska and Utah reported the lowest rates (116 and 105, respectively) and Florida reported the highest rate (267). ALS prevalence rates per 100,000 people with data up to the year 2015 are reported in separate quartiles of the country, i.e., averaging states in the Midwestern (5.5), Northeastern (5.1), Southern (4.7), and Western (4.4) regions (Mehta, 2018). Recently published ALS data updated to 2018 show prevalence rates averaging 4.4 (4.3, 4.5; 95 % confidence level, CI) per state and ranging from 2.6 per 100,000 people at the lowest in Hawaii (1.8, 3.3; 95 % CI) and Nevada (2.1, 3.2; 95 % CI) and 7.8 per 100,000 people at the highest in Vermont (5.9, 9.8; 95 % CI) (Mehta et al., 2024).

Despite its low prevalence, ALS, the debilitating and ultimately fatal neurodegenerative disease, is thought to progress due to apoptosis, necroptosis, pyroptosis, autophagy, or ferroptosis of motor neurons in the spinal cord and motor cortex of the brain (Nógrádi et al., 2024). Exposure to the cyanotoxin β -methylamino-L-alanine (BMAA) has been widely associated with ALS¹¹ as well as AD (Newell et al., 2022). The non-protein amino acid, BMAA, causes misfolding of proteins when inadvertently incorporated into protein formation (Myhre et al., 2013). Such malformed proteins in human neurons are known to form aggregates, leading to the development of neurofibrillary tangles (Myhre et al., 2013). Misfolded proteins, particularly with the Tau protein, are also involved in the pathology of AD and PD. Pathology of AD is characterized by β -amyloid ($A\beta$) plaques and misfolded Tau proteins, whereas patients with PD exhibit misfolded Tau protein and α -synuclein pathological inclusions (Gadhav et al., 2024). Sources of BMAA include bread wheat, cycad seeds, cyanobacteria, and diatoms, with worldwide detections at higher trophic levels in the food chain (e.g., in flying foxes) indicating species-dependent bioaccumulation (Li et al., 2024).

Exposure to BMAA is thereby possible through the ingestion of producers and consumers alike. Presumed means of exposure currently include ingestion of contaminated drinking water and food, inhalation of aerosolized BMAA, and human ingestion of supplements containing blue-green algae (Myhre et al., 2013; Li et al., 2024). Analysis of BMAA by LC-MS/MS typically is accomplished by extraction onto cation-exchange sorbents prior to elution and gradient separation on C₁₈ and HILIC columns using mixtures of water and acetonitrile as the mobile phase and formic acid as a modifier (Błaszczuk et al., 2021; Réveillon et al., 2014).

A toxicant more recently suggested to be associated with neurodegeneration is the polar herbicide glyphosate N-(phosphonomethyl) glycine, which has been shown to cross the blood-brain barrier and promote neuroinflammation (Winstone et al., 2022). Widespread use of glyphosate facilitates a spectrum of potential exposure routes, including ingestion of contaminated food and inhalation of aerosolized glyphosate (Ghisi et al., 2016). Glyphosate and its metabolite, aminomethylphosphonic acid or AMPA, have also been quantified successfully in animal food products using primarily LC-MS/MS methods (Verdini and Pecorelli, 2022). Previous methods for the detection of AMPA relied on the detection of target ions at m/z 110, 79 and 63 in negative ionization mode (Campbell et al., 2022; Amberger et al., 2023).

Wastewater-based epidemiology (WBE) today is a popular tool for inexpensive surveillance of populations large and small to environmental hazards including chemical and biological agents. Established methods have been utilized to monitor opioids (Castiglioni et al., 2006; Gushgari et al., 2019) and viruses such as SARS-CoV-2 (Ahmed et al., 2020; Bowes et al., 2022) in the wastewater of participating municipalities. WBE has been shown to be an innovative and efficient tool for monitoring indicators of public health due to its cost-effective, near real-time, noninvasive, and inclusive measures to monitor susceptibility to disease at the population level (Venkatesan and Halden, 2014; Westerhoff et al., 2015; He et al., 2016; Gil-Solsona et al., 2021; Kasprzyk-Hordern et al., 2023). Since exposure to BMAA and glyphosate have been linked to neurodegeneration, identifying regions with higher concentrations of these toxicants or their metabolites would signal the need for healthcare resources to be deployed to populations for further clinical screening. Detection of AMPA would indicate exposure to glyphosate and not just the occurrence of this toxicant circulating through the environment. The development of a WBE pipeline to detect chemical risk factors and biomarkers (Sharma et al., 2023) of NDDs represents a novel and potentially promising pathway to better monitor and manage the risk and prevalence of prominent NDDs.

The present exploratory study was motivated by preliminary data showing positive relationships between algal bloom occurrences and ALS prevalence rates in the US by region. In addition, the identification of a known metabolite from human exposure to glyphosate prompted the inclusion of AMPA in method development. In this first study to analyze BMAA levels in wastewater, we aimed to further examine potential associations between wastewater-borne chemical risk factors of NDD and prevalence rates of three major NDDs in the US geospatially by region and city.

2. Methods

2.1. Epidemiological data curation

Prevalence rates for ALS (Mehta, 2018; Mehta et al., 2024), PD (Marras et al., 2018), and AD (Alzheimer's, 2023) reported to the Centers for Disease Control and Prevention (CDC) were separately collected, converted to occurrences per 100,000 people, and combined into a generalized NDD prevalence rate for each geographical region. Algal bloom reports across the United States spanning the years of 2010 to 2020 were compared with regional and state-level prevalence rates of ALS (EWP, 2023). Glyphosate use reported to the US Geological Survey (USGS) in a 2019 preliminary report were compared with combined NDD prevalence rates. The percentage of drinking water sourced from surface water vulnerable to algal blooms was collected from city or state-level agencies responsible for regulatory reports.

2.2. Chemicals and reagents

Authentic chemical standard and labeled isotopes were purchased from Sigma Aldrich (St. Louis, MO) and Cambridge Isotope Laboratories (Tewksbury, MA), respectively. Analysis-grade methanol, acetonitrile, and purified water were purchased from Fisher Scientific (Thermo Fisher Scientific, Waltham, MA). Formic acid was purchased from Fluka Chemical Corp (Milwaukee, WI). Stock solutions prepared from target analytes and isotope-labeled analogs were stored in glass amber vials with polytetrafluoroethylene (PTFE) septa at 4 °C.

2.3. Sample selection

Wastewater samples from various municipalities around the United States ($n = 87$) received by the Human Health Observatory (HHO) within the Biodesign Center for Environmental Health Engineering were archived at -20 °C until further processing. Raw wastewater samplers collected composite samples within 24-h collection periods from influent wastewater not yet treated by the downstream wastewater treatment plant. Untreated wastewater samples therefore contained relatively fresh excrement from individuals contributing to the municipal pipeline. Samples were strategically selected to represent a variety of regions previously reported with high and low prevalence rates of NDDs and a targeted seasonal time window from March through October. Wastewater samples could therefore be compared spatially as well as longitudinally to observe analyte concentrations before, during, and after the summer (algal bloom) season between regions with historically high or low prevalence to NDDs.

2.4. Wastewater processing

Raw wastewater samples were thawed and pipetted into 100-mL aliquots in duplicates. Isotope-labeled internal standards were spiked into samples prior to solid phase extraction (SPE). Additionally, replicate samples were prepared in 300 mL and 600 mL aliquots for further AMPA analysis. A Strata-X-CW weak cation exchange (WCX) SPE cartridge (Phenomenex, Torrance, CA) was observed to yield better performance than use of an HLB cartridge for both BMAA and AMPA, as both analytes are polar compounds. Cartridges were eluted with LC-grade methanol containing 5 % formic acid to a final volume of 2 mL and stored in -20 °C refrigeration. Elutions were dried down and reconstituted in 10 % acetonitrile with 0.2 % formic acid in LC-grade water for LC-MS/MS analysis.

2.5. LC-MS/MS analysis

A Shimadzu LCMS 8060 (Columbia, Maryland) was utilized to analyze BMAA and AMPA by liquid chromatography tandem mass spectrometry (LC-MS/MS) with a UPHLC Raptor Biphenyl $5 \times 2.1\text{ mm}$

guard column for separation (Supplemental Table S1). BMAA and its isotope-labeled analog (BMAA- $^{13}\text{C}_3$) were run for 2.1 min with 90 % mobile phase A (100 % LC-grade water plus 0.2 % formic acid) and 10 % mobile phase B (100 % LC-grade acetonitrile plus 0.1 % formic acid) at a 0.4 mL/min flow rate. AMPA and its labeled analog (AMPA $^{13}\text{C}^{15}\text{ND}_2$) were run for 0.5 min with 90 % mobile phase A (100 % LC-grade water plus 0.2 % formic acid) and 10 % mobile phase B (100 % LC-grade acetonitrile plus 0.1 % formic acid) at a 1.5 mL/min flow rate. Peaks were integrated using the Shimadzu software Lab Solutions Insight.

2.6. Quality assurance

BMAA ($m/z\ 119.2 \rightarrow 102.2$ (91.2) and AMPA ($m/z\ 110.0 \rightarrow 79.0$ (61.1)) parent and product targets (quantitative and qualitative targets respectively) were identified using MRM optimization. The BMAA standard curve was developed with 13 data points and an R_{sq} (R -squared linear regression statistic) value of 0.999. The AMPA standard curve was developed with 10 data points and an R_{sq} value of 0.993. Negative controls in the form of processing blanks (DI water) and analytical blanks (mobile phase media) were utilized in every chemical step and after a maximum of 10 samples during LC-MS/MS analysis.

2.7. Statistical analysis

Statistical significance of associations between algal bloom reports and prevalence rates was determined using the likelihood ratio by contingency analysis. Significance between wastewater concentrations and prevalence rates was determined through one-way analysis of variance and reporting of p -values (Gao et al., 2023a, 2023b; Gao et al., 2024). The Student's t -test (Wang et al., 2024a; Wang et al., 2024b) was used to test the significance of the correlation between reported algal report blooms and wastewater concentration.

3. Results

In this exploratory study, sampling locations for municipal wastewater previously established by the Human Health Observatory at Arizona State University were employed to study reported algal blooms between the years 2010 and 2020 in the context of ALS prevalence rates; regions reporting algal blooms showed higher ALS prevalence rates per 100,000 people by quartile ($p < 0.0001$) (Supplemental Fig. S1). When state-level data was available and used, the significance of the correlation persisted ($p = 0.01$) (Fig. 1a). Analyses were conducted by binning cities and states based on the presence or absence of algal blooms. Cities with the highest ALS prevalence rates by quartile (>5.0 per 100,000 annually) also reported algal blooms, i.e., Traverse City and Brownstown in Michigan; Dodge City, Kansas; West Chicago and Carol Stream in Illinois; and Hoboken, New Jersey. Of the regions with the lowest ALS prevalence rate (4.4) nationally, nine cities within six states did not report algal blooms since 2010. However, 17 cities within eight states reported the lowest ALS prevalence rate as well as algal blooms. Meanwhile, no relationship was found between reports of algal blooms, percentage of surface water used for drinking water, and ALS prevalence rates by quartile ($p = 0.71$) or state ($p = 0.15$) (Supplemental Fig. S1, Fig. 1b). As for the comparison of NDD prevalence rates and glyphosate use reported to the USGS by region, no correlation was found (Supplemental Figs. S6 and S7). States with higher NDD prevalence rates (e.g., Florida, Arizona, New Jersey, and New Mexico) widely varied in their glyphosate use level, but skewed more toward low to moderate use. Meanwhile, states with low NDD prevalence rates (e.g., Alaska, Texas, Colorado, Georgia, and Idaho) reported primarily high to very high glyphosate use levels. This trend shows either a lack of a correlation or possibly a negative correlation between the prevalence of NDDs and glyphosate use, albeit insignificant.

Upon method development, the BMAA limit of detection (LOD) was determined to be 0.0075 ppb, when using ten-times the signal noise

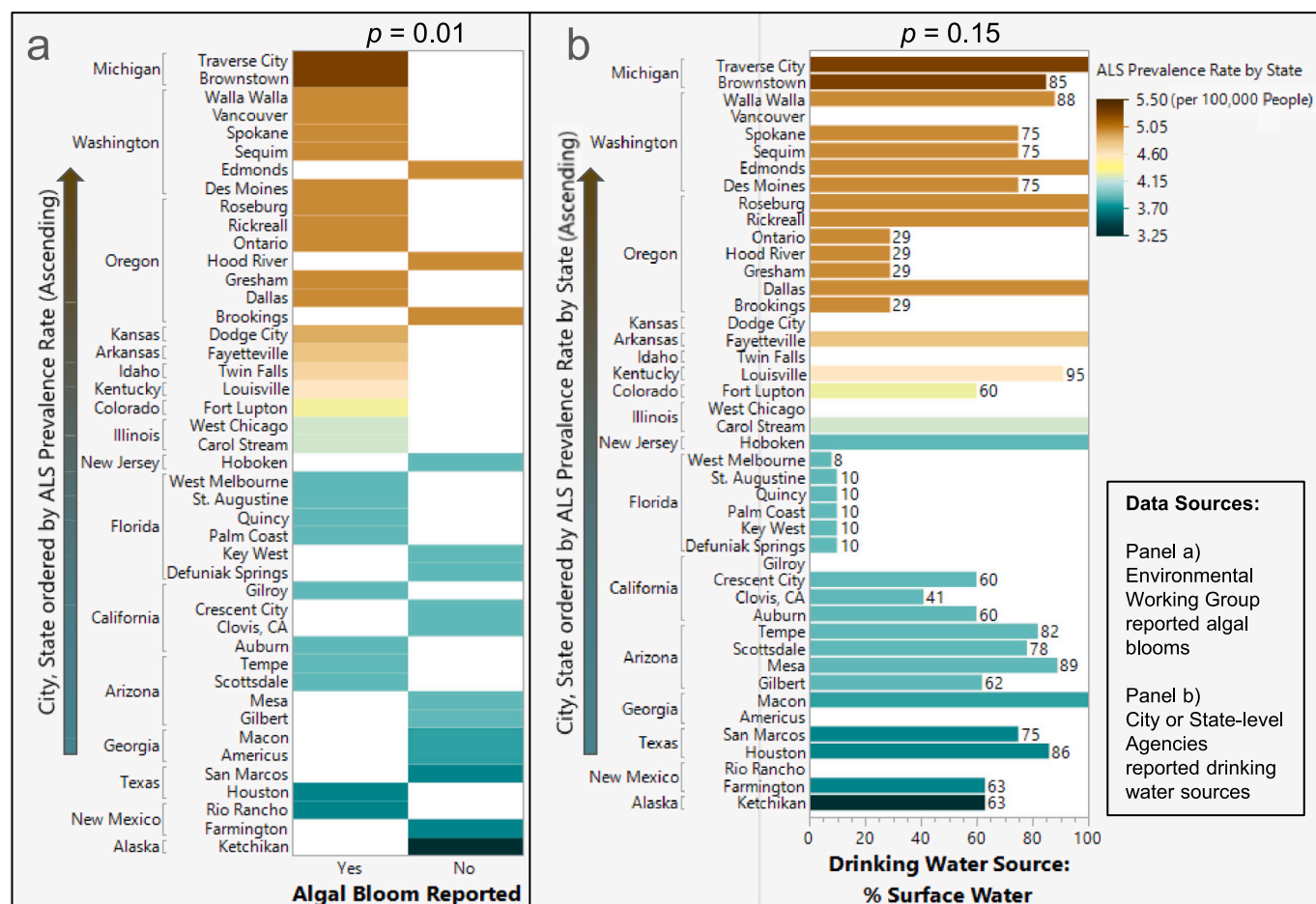


Fig. 1. Geographic Distribution of Reported Algal Blooms Ranked by Amyotrophic Lateral Sclerosis Prevalence Rates by State in Selected Cities within the United States. Bar charts show sources of β -methylamino-L-alanine exposure from consumption of surface water experiencing algal blooms. a) Algal bloom presence reported by the Environmental Working Group by city from 2010 to 2020 and b) percent drinking water sourced by surface water reported by city are ranked by amyotrophic lateral sclerosis (ALS) prevalence rates per 100,000 people by state. A heat map is utilized to indicate states with lower ALS prevalence rates (green) compared to states with higher ALS prevalence rates (brown). Cities included are within the Human Health Observatory network. P values (with 95 % Confidence Intervals) are given to show the relationship between a) algal blooms by ALS prevalence rates and b) surface water percent by ALS prevalence rates.

surrounding the detected peak as the detection threshold. The limit of quantification (LOQ), however, was determined to be 0.01 ppb (Supplemental Fig. S3) by finding the minimal concentration suitable to reproducibly obtain a standard curve with an R_{sq} value of 0.999 or greater. In contrast, AMPA showed a comparatively higher LOQ of 4.07 ppb and overall poorer analytical performance (Supplemental Fig. S8). Additionally, attempting to concentrate high-volume samples of up to 600 mL also resulted in no detection of AMPA in wastewater from any region sampled in the HHO network upon LC-MS/MS analysis.

Fig. 2 shows the various locations around the United States that were sampled for wastewater and analyzed for BMAA concentrations (Fig. 2c, Supplemental Table S2). Availability of ALS prevalence rate data in the literature at the time of experimental design drove sampling location choices within regions (quartiles shown in Fig. 2a) with a variety of ALS prevalence rates. The number of samples within each quartile was driven by accessibility within the sampling network during the 2022 sampling period, showing a larger number of samples in Arizona and Washington states compared to other regions across the United States. Recently published ALS prevalence rates by state were mapped (Fig. 2b) for post-run analysis, indicating more variability at a more highly resolved level. For instance, prevalence rates for Washington state and Oregon were higher than what was observed at the quartile level (5.0 compared to 4.4 per 100,000 people for both), while rates for Illinois and New Jersey were lower than reported by quartile (4.2 and 3.9 compared to 5.5 and 5.1 per 100,000 people, respectively).

Analysis of wastewater from regions within the sampling network showed an insignificant correlation ($p = 0.18$) between reported algal blooms and BMAA concentrations (Fig. 3a, Supplemental Fig. S4). However, BMAA concentrations were significantly correlated ($p < 0.0001$) with ALS prevalence rates (Fig. 3b, Supplemental Fig. S4). Specifically, BMAA concentrations in wastewater were highest on average (1.86 ppb) in regions with the highest ALS prevalence rate (5.5). Although the lowest BMAA concentration on average (0.04 ppb), however, was not seen in the lowest ALS prevalence rate quartile, the second to the highest (5.1) was in fact observed in regions with the lowest prevalence. While BMAA concentrations in wastewater from regions in the middle quartiles were observed to be the lowest, the average concentration (0.44 ppb) in the lowest quartile (4.41) was still significantly lower than what was observed on average in the highest quartile.

Potential seasonal effects were examined by comparing BMAA concentration and sample date. The LC-MS/MS method developed for BMAA was applied to wastewater samples across the US, aiming to capture conditions before, during, and after algal bloom events. Higher concentrations of BMAA were observed between late May and June of 2022 (Fig. 4, Supplemental Fig. S5), with the exception of a sample from Carol Stream, Illinois, which showed an elevated average BMAA concentration (4.67 ppb) detected March 13, 2022 (Supplemental Figure); during curve fitting, this data replicate set was treated as an outlier, as its positioning on the fringe of the data set otherwise would have been moderated only by one adjacent observation and not bilaterally by two

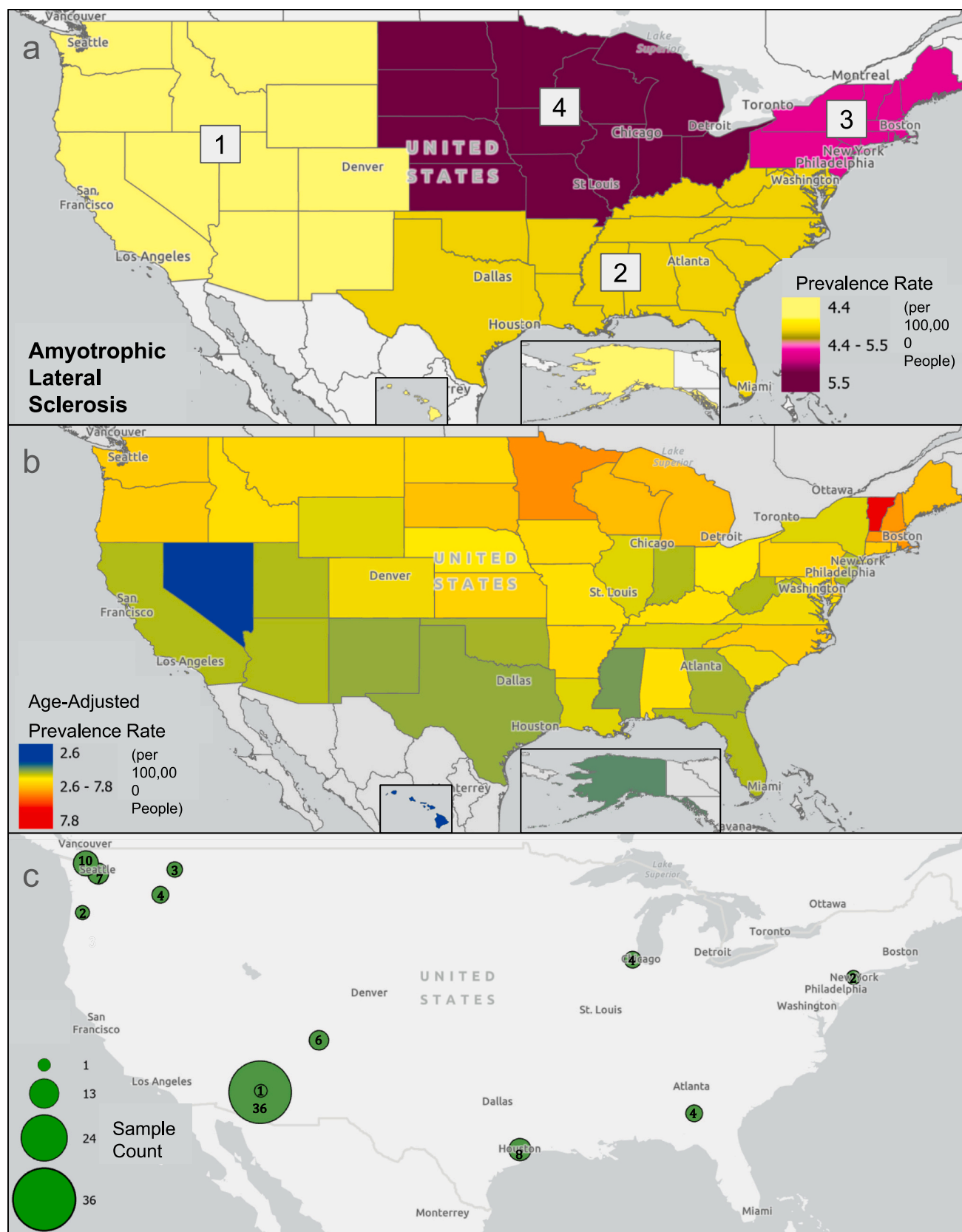


Fig. 2. Maps of Amyotrophic Lateral Sclerosis Prevalence Rates and Wastewater Sample Count Across the United States. Wastewater sample selection was aided by identifying regions within the Human Health Observatory network that varied by prevalence rates of amyotrophic lateral sclerosis (ALS). ALS prevalence reported per 100,000 people a) by quartile (Mehta, 2018) and b) later updated by state (Mehta et al., 2024) are mapped in ArcGIS with color gradients: lowest prevalence rates in blue and highest prevalence rates in red. c) Wastewater samples ($n = 87$) sourced from the Human Health Observatory analyzed for BMAA concentration are mapped in ArcGIS with sample count indicated by size.

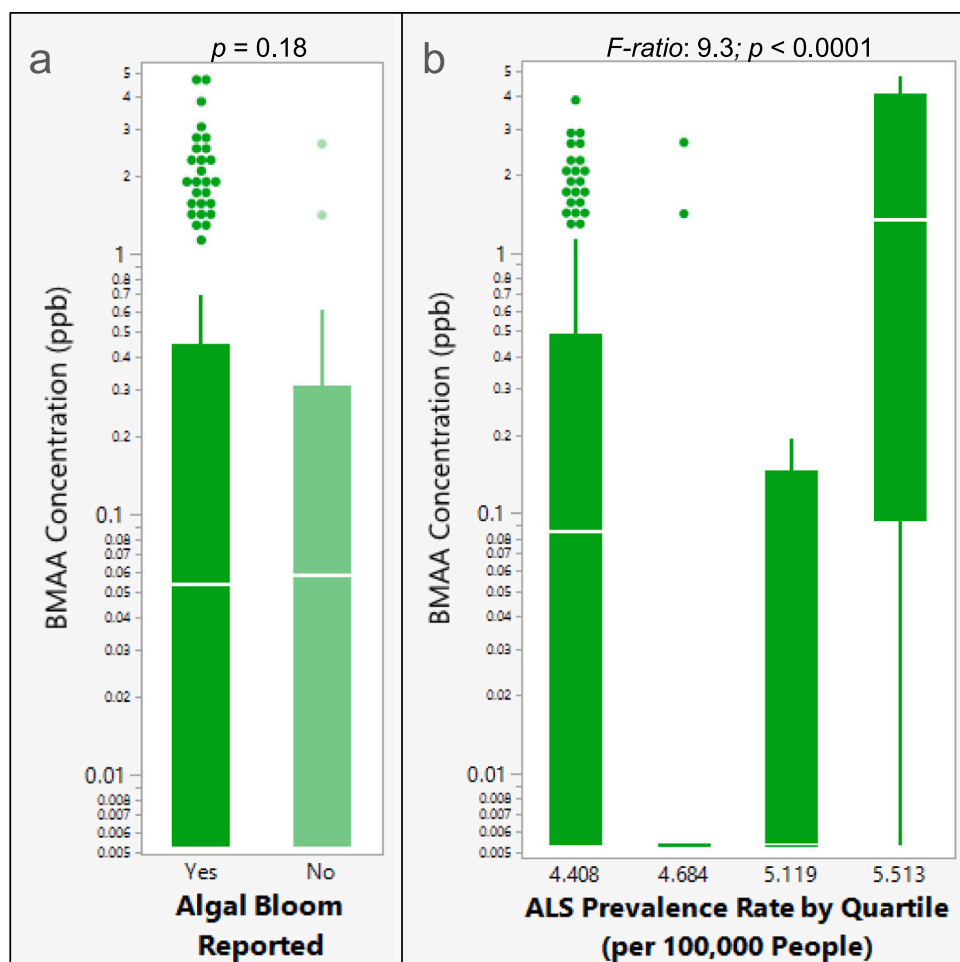


Fig. 3. Wastewater BMAA Concentration by Algal Bloom Reports and Amyotrophic Lateral Sclerosis Prevalence Rates. Levels of β -methylamino-L-alanine (BMAA) quantified in wastewater using LC-MS/MS were compared to reported algal bloom and amyotrophic lateral sclerosis (ALS) prevalence rates to determine if concentrations were correlated with these previously reported epidemiological data. Average BMAA concentration (ppb) of wastewater samples ($n = 87$) collected between March and October 2022 plotted on a logarithmic scale by a) algal bloom reports in the same city reported by the Environmental Working Group from 2010 to 2020 and b) ALS prevalence rate per 100,000 people (Mehta, 2018) in the same quartile. P values (with 95 % Confidence Intervals) are given to show the relationship between BMAA concentration and a) algal bloom reports in a Student's t -test and b) ALS prevalence rates by quartile in an analysis of variance using an F-ratio comparing variance between quartiles and within quartiles.

observations (Fig. 4 shows the curve without the pseudo surge present at the beginning of the time range of the study, and inclusion of the data point resulted in the curve shown in the Supplemental Fig. S5a). This sample was determined to be an outlier and therefore omitted in the curve fit due to the five times higher BMAA concentration average of replicates compared to other sample readings from the same sample time window. Nevertheless, statistical analysis was performed with all data, including the data point removed for curve fitting, and still indicated a highly significant probability ($p = 0.0009$) of BMAA concentrations to be higher during the summer months (i.e., May and June) compared to the rest of the sample date window (Supplemental Fig. S5b).

4. Discussion

4.1. BMAA in wastewater

The hypothesis that ALS prevalence rates are higher in geographic regions reporting algal blooms was supported by both regional and state-level analyses. Locations within the US reporting algal blooms were significantly elevated in ALS prevalence rates by geographic region (Likelihood Ratio 27.52; $p < 0.0001$) and by state ($p = 0.01$) (Supplemental Fig. S1, Fig. 1a).

However, no relationship was found between ALS prevalence rates and locations reporting algal blooms and drawing surface water for drinking water (Fig. 1b and Supplemental Fig. S1). Even though our team was able to find city-level information regarding the percentage of drinking water sourced from surface water, similar ALS prevalence rates resolved at the city-level were not available as it is reported only at the regional or state level. Had city-level prevalence rate data been available for neurodegenerative diseases, then a correlation may still have been observable, but this was not possible here. Restricted data limit the ability for this study to test the hypothesis at a more highly resolved level than region and state.

When this research study was planned and conducted, ALS prevalence rates reported to the CDC and available to this research study provided regional data (i.e., midwestern, northeastern, southern, and western) (Fig. 2a), while PD and AD rates were reported by state. Future work would benefit from matching wastewater sampling locations from states where higher resolution NDD data is available (Fig. 2b).

BMAA concentrations found in wastewater were also correlated ($p < 0.0001$) with the lowest (4.4) and highest (5.5) ALS prevalence rates per 100,000 people (Fig. 3b). However, reports of algal blooms were found to not be significantly correlated ($p = 0.18$) with BMAA concentrations in wastewater from locations sampled (Fig. 3a). As algal blooms are known to be seasonal, we anticipated higher concentrations of

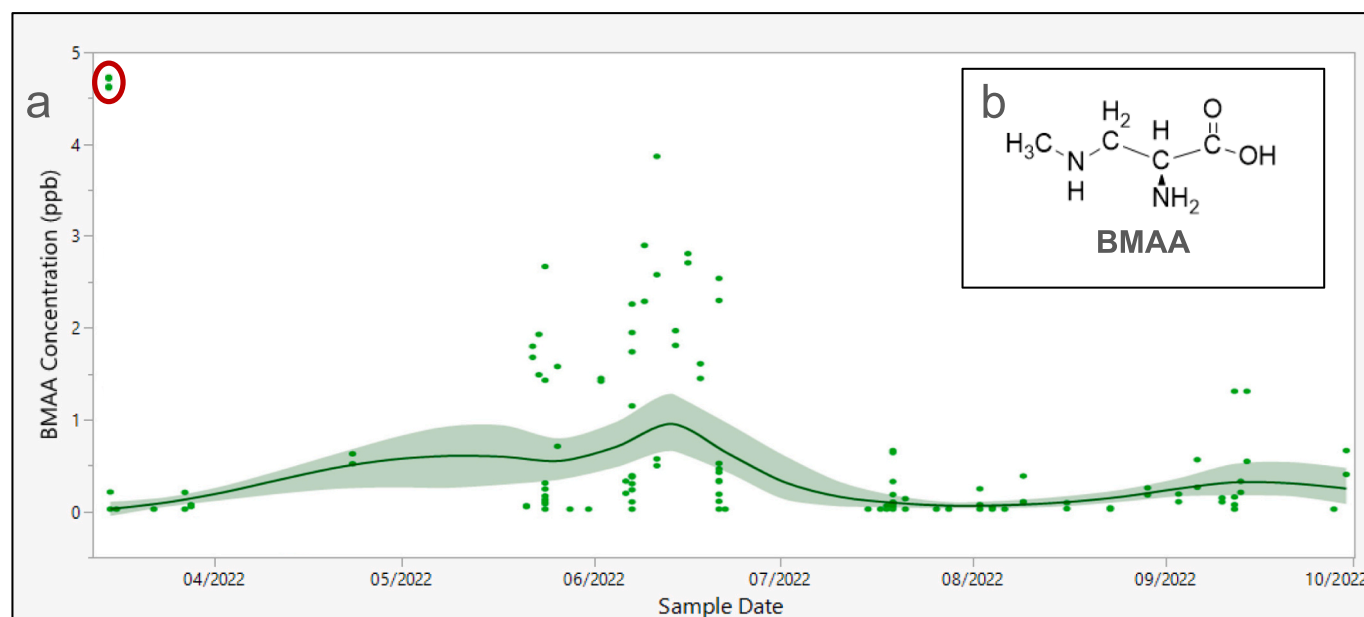


Fig. 4. Concentration of BMAA in Wastewater by Sample Date. To determine if a time-dependent relationship occurs between β -methylamino-L-alanine (BMAA) concentrations in wastewater and seasonality, BMAA levels were plotted over the sample date window. a) Average BMAA concentration (ppb) of wastewater ($n = 87$) sampled between March and October 2022 is plotted in a scatter plot by sample date. Curve fit uses piecewise cubic splines with confidence of fit shading. The red circle indicates the data for Carol Stream, Illinois from March 13, 2022, that was removed from analysis. Raw analysis including outlier data was treated in Supplemental Fig. S5. b) The chemical structure of β -Methylamino-L-alanine (BMAA).

BMAA in the summer months compared to the Spring and Fall. Indeed, most wastewater samples containing detectable levels of BMAA were taken between the end of May and the beginning of August (Fig. 4). However, one sample in March of 2022 taken in Carol Stream, Illinois showed an unexpectedly high concentration of BMAA relative to other samples in the time period prior to the summer months from this location. Upon further inspection, water supplied to Carol Stream, just west of Chicago, is from Lake Michigan. Surface water temperatures in Lake Michigan on the coast of Chicago was reportedly 1.9 °C in the month of March of 2022 (Seatemperature, 2024). Since this temperature is much colder than what is needed for algal blooming, it is unlikely that the detected BMAA resulted from seasonal cyanobacterial growth. If BMAA concentrations were indeed high in Lake Michigan in the month of March, other sources of BMAA must be considered (e.g., organisms with bioaccumulated BMAA).

4.2. AMPA and glyphosate

Prevalence rates of NDD did not correlate with glyphosate use reported to the USGS. In fact, the regions with the lowest NDD prevalence rates reported the highest glyphosate use. While glyphosate has been significantly associated with neurodegeneration, this finding was unexpected and here remains unexplained.

All samples tested for AMPA yielded results of non-detect at an LOQ of 4.07 ppb, which was higher than that for BMAA. Additional steps may be required to further increase the sensitivity of the LC-MS/MS method, including the use of higher volume samples, longer gradient runs, use of longer analytical columns or even additional sample pre-concentration steps. In summary, additional preparatory steps likely will be necessary for quantification of AMPA with higher sensitivity at the sub-ppb level.

4.3. Wastewater-based epidemiology

The current WBE methodology primarily targets markers occurring in the environment rather than evidence of exposure. Known and putative biomarkers, including metabolites that are released from the body

as a result of exposure to NDD toxins, could provide an early-warning signal of harmful environmental conditions or human behaviors resulting in risky exposures to NDD-associated agents (Fig. 5). Leveraging wastewater as an observation point of combined exposure of a community would expand the monitoring of susceptibility from the individual to the population level. While BMAA does not have a known metabolite and therefore can only be quantified from an occurrence standpoint (Downing et al., 2019), developing a method to quantify AMPA as a metabolite would offer epidemiologists information about exposure to the toxic chemical glyphosate. A method to target a metabolite such as AMPA would need to be developed with a limit of detection capable of quantifying the compound when present at sub-ppb levels in the complex wastewater matrix. Ultimately, a signal with the highest possible intensity is needed as well as a high dynamic range of said signal to trace one or multiple NDDs which feature relatively low prevalence rates in human populations.

4.4. Potential limitations

Time is an important factor to consider in epidemiological studies. While environmental exposure to these organic compounds have been significantly associated with NNDs, dose and time effects may limit the value of measuring current occurrence of chemical risk factors and exposure biomarkers in the environment today. Our study found the presence of BMAA in wastewater collected in the summer of 2022 was significantly correlated with algal report blooms since 2010. However, animal studies have suggested exposure to BMAA is more toxic to humans and therefore will lead to more aggressive progression of neurodegeneration under longer time spans even when the doses are relatively low (Campbell et al., 2022).

Concentrations of target compounds may also be impacted and modified by metabolic reactions in raw wastewater, which contains a multitude of active microorganisms (Gao et al., 2023a, 2023b). Collecting and preserving wastewater (e.g., applying salt buffers and refrigeration) closer to the source (e.g., building-level cleanouts or neighborhood manhole) rather than a downstream sampler entering the wastewater treatment plant would reduce the possibility of biological

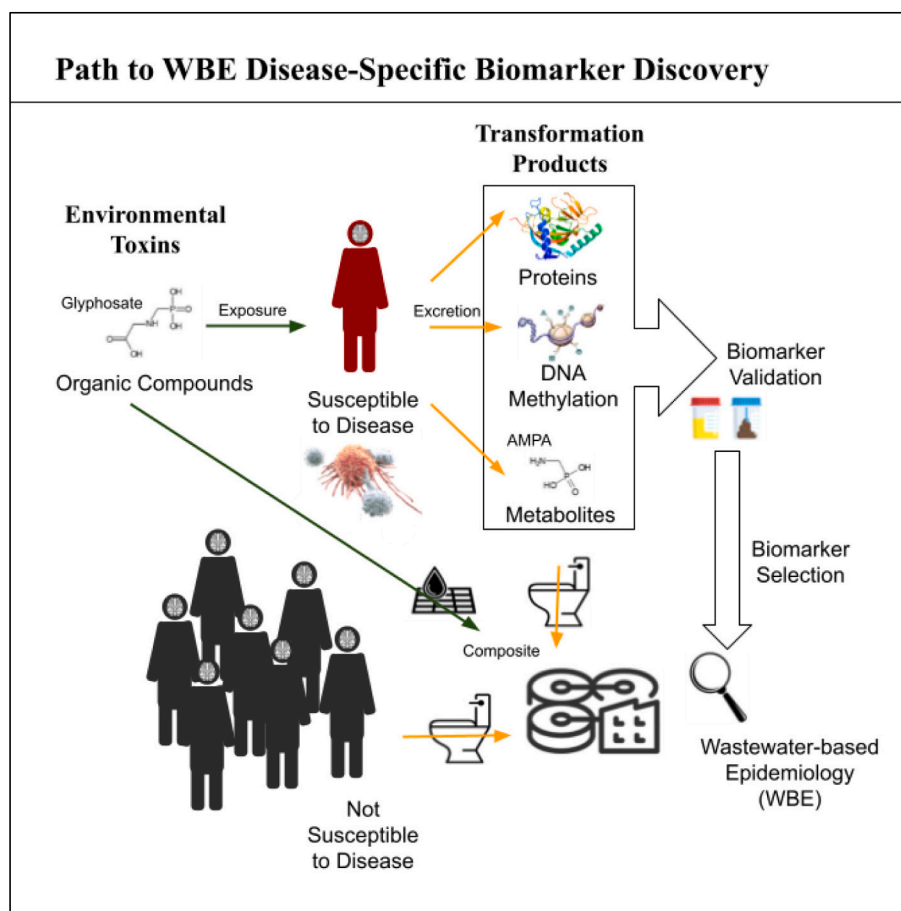


Fig. 5. Discovering Disease-Specific Biomarkers through Wastewater-Based Epidemiology. A potential wastewater-based epidemiology (WBE) path to discover targeted disease-specific biomarkers is charted from the source of environmental toxins to biomarker types produced by individuals that could be targeted in wastewater analysis at community-level. Exposure to organic compounds leads to an individual excreting transformation products validated as biomarkers by epidemiological work in clinical practice. WBE methodology selects biomarkers to monitor in composite wastewater samples to determine populations susceptible to disease.

degradation. Moreover, target compounds may degrade over time in legacy wastewater samples stored under freezing conditions (Gao et al., 2023a, 2023b). Chemical degradation from both long-term refrigeration and potential interferences from utilizing raw wastewater as a complex matrix could artificially reduce signal intensities detected on the LC-MS/MS as samples remain archived in perpetuity.

Additionally, the HHO network provides access to legacy samples across many states in the US, but it is not comprehensive. Limited access to both wastewater samples and matching NDD data of high geospatial resolution were impairing the present study. Inconsistent results may indeed have been observed from low geospatial resolution and potential confounders (e.g., higher limits of detection, matrix interferences, the lack of algal blooms during the sample period, etc.). Future work would benefit from continued and expanded collaboration with municipalities across additional states.

4.5. Concluding thoughts

This study aimed to determine the utility of wastewater-borne BMAA and AMPA as markers of NDD prevalence. WBE is a budding field that is capable of tracking indicators of public health beyond illicit drug use or infectious disease outbreaks. The proof of concept provided here with this LC-MS/MS method for quantifying BMAA in wastewater presents a novel approach to using WBE beyond acute and infectious disease applications. Key findings include significant correlations between algal blooms reported as well as higher BMAA concentrations, particularly in

the summer months, in regions with ALS prevalence rates above the mean compared to regions below the mean. This study hereby promotes a continuation of this line of research, with both toxicants occurring in the environment and toxicants that metabolize in the human body, because it represents an innovative approach to monitor for diseases with immediate, intermediate and even delayed onset, with the latter including a spectrum of chronic neurological disorders. Thus, WBE appears primed to continue its evolution as a broadly applicable, inexpensive, informative, and therefore invaluable tool for public health monitoring and management.

CRediT authorship contribution statement

Melanie Engstrom Newell: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ayesha Babbrah:** Writing – review & editing, Software, Methodology, Investigation, Conceptualization. **Anumitha Aravindan:** Writing – review & editing, Software, Methodology, Investigation, Conceptualization. **Sharvaree Kulkarni:** Writing – review & editing, Software, Investigation, Formal analysis, Data curation. **Alexandria Ellershaw:** Writing – review & editing, Software, Investigation, Formal analysis, Data curation. **Aashika Dupati:** Writing – review & editing, Software, Investigation, Formal analysis, Data curation. **Raj Rathnam:** Writing – review & editing, Conceptualization. **Gray Shaffer:** Writing – review & editing, Data curation. **Lesley Estrada:** Writing – review & editing, Data

curation. **Courtney Curtis:** Writing – review & editing, Data curation. **Justin Leneaux:** Writing – review & editing, Data curation. **Erin M. Driver:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Rolf U. Halden:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Rolf U. Halden reports financial support was provided by Glen Swette Memorial Fund. All other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2025.179032>.

Data availability

Provided in Supplemental Material

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Glossary

Wastewater-based epidemiology (WBE): An approach to identifying and monitoring

indicators of public health by analyzing composite wastewater and sewage sludge media.

Amyotrophic lateral sclerosis (ALS): A neurodegenerative disease affecting motoneurons, also known as Lou Gehrig's Disease.

Prevalence rates: The number of people in a population who have been diagnosed with a disease within the time frame (e.g., year).

β -methylamino-L-alanine (BMAA): A non-proteinaceous amino acid cyanotoxin, associated with neurodegeneration upon human exposure.

Aminomethylphosphonic Acid (AMPA): A metabolite of the herbicide glyphosate indicator human exposure, associated with neurodegeneration.

Liquid chromatography tandem mass spectroscopy (LC-MS/MS): An analytical technique combining the separation from a column and the detection of target compounds by mass.

Algal blooms: Excessive algal growth in waterways occurring in a short period of time, a phenomena known as eutrophication.